

Amman Patel - Master Resume

Los Angeles, CA | amman.patel11@gmail.com | (949)-207-8111 | <https://www.linkedin.com/in/amman-patel/> | U.S. Citizen

Education

University of California, Riverside | BS, Mechanical Engineering

June 2026

- Federal Grant, Middle Class Scholarship, UCR Grant

Tesoro High School | High School Diploma

June 2021

- Graduated Valedictorian (Ranked #1 out of 250 students)

Clubs and Projects

BCI Wheelchair (Fusion 360, Python, SolidWorks, mindaffectBCI)

January 2026 – May 2026

- Engineered a non-invasive, BCI-controlled wheelchair system to reduce reliance on joystick and sip-and-puff control, with the goal of enabling users with advanced neuromuscular disorders or limb loss to navigate via visual stimuli.
- Designed and jointly iterated on a custom linear actuator mount to steer the chair within $\pm 2\%$ precision.
- Tested the accuracy and delay of live EEG signals acquired from a 4-channel OpenBCI headset after using BrainFlow to stream and process those signals in Python.
- Worked with mindaffectBCI to develop a cVEP stimulus matrix and decoding algorithm, troubleshooting signal classification issues to translate raw EEG focus into 5 directional commands, achieving 0.95 s end-to-end response latency (target: < 2.0 s) and 98% classification accuracy (target: $> 85\%$).
- Evaluated various VEP methods (P300, SSVEP, cVEP) through research to select the best-suited method for response speed and reliability.
- Led outreach to industry contacts, including Invacare and prior teams in assistive BCI systems, to inform design requirements and troubleshoot mechanical integration challenges.
- Applied a weighted decision matrix across 6 criteria to evaluate 4 candidate concepts and justify final design selection.
- Won first prize among the 15 teams.

Highlander Space Program: Project Poseidon (ANSYS/Fusion 360/Open Rocket/SolidWorks Flow Simulation)

August 2023 –

Purpose: To design a bipropellant rocket capable of depositing native seeds over a large field.

September 2024

- Contributed to the design and analysis and balanced aerodynamic performance, structural integrity, and material constraints of a 120 in long, 7 in diameter bipropellant rocket (87.6 lb. fully loaded) planned for large-area seed dispersal.
- Examined internal and external fluid behavior through combustion and flow (CFD) analyses and altered airframe and fin designs based on these simulation results, achieving a stable flight profile with a 2.79 caliber stability margin (CG at 69.356 in, CP at 88.879 in) and a predicted apogee of 35,930 ft.
- Iteratively assessed fin configurations with varying parameters to minimize induced drag and thermal loading whilst maintaining stable airflow characteristics, supporting a simulated max velocity of 1,582 ft/s (Mach 1.51) and max acceleration of 5.77 G.
- Researched and coordinated with suppliers of airframe material candidates, including, but not limited to aluminum, carbon fiber, steel, and custom composites, ultimately selecting a processed fiber composite airframe with PLA-printed nose-to-body connectors; used Young's Modulus to perform hand calculations to test the strength and loading limits of these materials under launch conditions, achieving a FOS of 3 on the nose cone.

Vacuum Design Project (SolidWorks)

October 2022 – December 2022

- Created a vacuum to clean solar panels attached to the roof of the average American home that could be taken apart and assembled by the consumer.
- Collaborated with peers using the 6-3-5 brainwriting method to generate concepts, narrowing down to three finalists before individual detailed design development.
- Sketched preliminary designs and selected the design that was relatively simple to manufacture while still being effective for the vacuum's intended purpose.
- Performed FEA to stress-test the primary CAD model, iterating on dimensions and material selection to reach a finalized design.
- Composed an engineering document, digestible to both non-technical and technical readers, examining the conceptual designs in a Pugh matrix, the utility of the product, an exploded view of the final design, etc.

Professional Experience

UPS, Associate

July 2022 – September 2022

- Assisted +100/day customers with shipping and returning packages, print files, fax documents, etc.
- Checked in 55-75/day incoming packages from UPS drivers and consolidated return packages into large boxes efficiently
- Investigated various issues, usually relating to customer pick-ups, and coordinated with coworkers to find and address these issues, leading to double the productivity.

beyondGreen Biotech Inc, Intern

June 2020 – August 2020

- Assessed and corresponded with 17 potential suppliers of coffee waste for biodegradable plastic.
- Oversaw and managed the packaging of various products made with biodegradable plastic, such as hardware
- Operated 4 specialized machines, in collaboration with 7 team members, to process, melt, mold, and set coffee grounds as well as other sources of cellulose into usable products.
- Maintained the company website to modernize and add an e-catalog and e-commerce components

Technical Skills

CAD/Simulation: ANSYS, SolidWorks (CSWA Certified December 24, 2021), SOLIDWORKS Flow Simulation, Fusion 360, Open Rocket, and mindaffectBCI

Programming: MATLAB and Python